Tutorial - 2

(1) $L = \frac{1}{2} m\dot{x}^2 - \frac{1}{2} kx^2$

$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$

(i) $H = \sum p_i \dot{q}_i - L$

$= p\dot{q} - L$ (1D motions.

$= \frac{\partial L}{\partial \dot{x}} \dot{q} - L$

$= m\dot{x}^2 - \frac{1}{2} m\dot{x}^2 + \frac{1}{2} kx^2 = \frac{1}{2} m\dot{x}^2 + \frac{1}{2} kx^2$ ~~(Ans)~~

(ii) $\dot{p} = -\frac{\partial H}{\partial x} = \cancel{m\dot{x}} - kx$ — (I)

$\boxed{H = \frac{p^2}{2m} + \frac{1}{2} kx^2}$ (Ans)

$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}$ — (II)

Differentiating (II) wrt time : $\ddot{x} = \frac{\dot{p}}{m} = -\frac{kx}{m}$

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$\Rightarrow$ $\boxed{m\ddot{x} + kx = 0}$ — Hamiltonian's combined EOM.

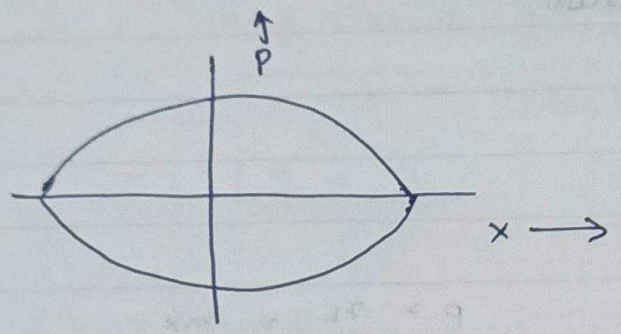
(iii) $m\ddot{x} + kx = 0$.

$\Rightarrow \ddot{x} + \frac{k}{m}x = 0$ $\Rightarrow \ddot{x} + \omega^2 x = 0$ $\left[\omega^2 = \frac{k}{m}\right]$.

General soln: $x(t) = A\cos\omega t + B\sin\omega t$ (Ans) (*)

(iv) $\frac{p^2}{2m} + \frac{1}{2}kx^2 = E$ (total energy).

$\Rightarrow$ $\boxed{\frac{p^2}{2mE} + \frac{x^2}{2E/k} = 1}$ — eqn of ellipse



(*) $\dot{x}(t) = -\omega A\sin\omega t + \omega B\cos\omega t$.

$\Rightarrow p = -m\omega A\sin\omega t + m\omega B\cos\omega t$. (Ans)

(2) $L = m\dot{x}\dot{y} + \frac{\gamma}{2}(x\dot{y} - \dot{x}y) - kxy$

(1) $\frac{\partial L}{\partial x}$ for x:

$\frac{\partial L}{\partial \dot{x}} = m\dot{y} - \frac{\gamma}{2}y$ $\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{y} - \frac{\gamma}{2}\dot{y}$.

$\frac{\partial L}{\partial x} = \frac{\gamma}{2}\dot{y} - ky$.

$\therefore$ $m\ddot{y} - \frac{\gamma}{2}\dot{y} - \frac{\gamma}{2}\dot{y} + ky = 0$ $\Rightarrow$ $\boxed{m\ddot{y} - \gamma\dot{y} + ky = 0}$

(Ans)

Date.....

Expt. No..... Page No.....

For y :

$$\frac{\partial L}{\partial y} = m\dot{x} + \frac{\gamma}{2}x \Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) = m\ddot{x} + \frac{\gamma}{2}\dot{x}$$

$$\frac{\partial L}{\partial y} = -\frac{\gamma}{2}\dot{x} - kx.$$

$$\therefore \quad m\ddot{x} + \frac{\gamma}{2}\dot{x} + \frac{\gamma}{2}\dot{x} + kx = 0.$$

$$\Rightarrow \quad \boxed{m\ddot{x} + \gamma\dot{x} + kx = 0} \quad \text{(Ans)}$$

(b) $H = \sum_i p_i \dot{q}_i - L$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{y} - \frac{\gamma}{2}y. \Rightarrow \dot{y} = \frac{1}{m}\left(p_x + \frac{\gamma}{2}y\right)$$

$$p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{x} + \frac{\gamma}{2}x \Rightarrow \dot{x} = \frac{1}{m}\left(p_y - \frac{\gamma}{2}x\right)$$

$$\therefore H = p_x\dot{x} + p_y\dot{y} - L$$

$$= m\dot{x}\dot{y} - \cancel{\frac{\gamma}{2}xy} + \cancel{m\dot{x}\dot{y}} + \cancel{\frac{\gamma}{2}x\dot{y}} - \cancel{m\dot{x}\dot{y}} - \frac{\gamma}{2}(\cancel{x\dot{y}} - \cancel{\dot{x}y}) + kxy.$$

$$H = m\dot{x}\dot{y} + kxy.$$

$$\boxed{H(q, p, t) = \frac{1}{m}\left(p_y - \frac{\gamma}{2}x\right)\left(p_x + \frac{\gamma}{2}y\right) + kxy} \quad \text{(Ans)}$$

~~(H)~~ ~~(a)~~ 3 (1) $\frac{d^2y}{dt^2} + \omega^2 y = k.$

CF: Solve for $\frac{d^2y}{dt^2} + \omega^2 y = 0.$

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$\therefore y_c = A \sin \omega t + B \cos \omega t$

Pr: Solve for $\frac{d^2y}{dt^2} + \omega^2 y = k$

Let $y_p = \frac{k}{\omega^2} \Rightarrow \frac{d^2y}{dt^2} = 0$

$\therefore y(t) = y_c + y_p = A \sin \omega t + B \cos \omega t + \frac{k}{\omega^2}$ (Ans)

(ii) $\frac{d^2y}{dx^2} + 2\gamma \frac{dy}{dx} + \omega^2 y = P \sin \Omega x$

$\Rightarrow y'' + 2\gamma y' + \omega^2 y = P \sin \Omega x$

CF:

$y'' + 2\gamma y' + \omega^2 y = 0$ — characteristic eqn

Roots: $r = \frac{-2\gamma \pm \sqrt{4\gamma^2 - 4\omega^2}}{2}$

$r = -\gamma \pm \sqrt{\gamma^2 - \omega^2}$

If $\gamma < \omega$: $\rightarrow r = -\gamma \pm ip$; $p = \sqrt{\omega^2 - \gamma^2}$

$y_c(x) = e^{-\gamma x}(c_1 \cos px + c_2 \sin px)$

If $\gamma = \omega$: $\rightarrow r = -\gamma, -\gamma$.

$y_c(x) = (c_1 + c_2 x) e^{-\gamma x}$

If $\gamma > \omega$: $\rightarrow r = -\gamma \pm q$; $q = \sqrt{\gamma^2 - \omega^2}$.

$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$; $r_1 = -\gamma + q$

$r_2 = -\gamma - q$

PI :

Let $y_p = a \sin(\Omega x) + b \cos(\Omega x)$

$\therefore y' = \Omega a \cos(\Omega x) - \Omega b \sin(\Omega x)$

$y'' = -\Omega^2 a \sin(\Omega x) - \Omega^2 b \cos(\Omega x)$

$= -\Omega^2 [a \sin(\Omega x) + b \cos(\Omega x)]$

Substituting into the eqn:

$$-\Omega^2 [a \sin(\Omega x) + b \cos(\Omega x)] + 2\gamma\Omega [a \cos(\Omega x) - b \sin(\Omega x)] + \omega^2 [a \sin(\Omega x) + b \cos(\Omega x)] = F \sin(\Omega x)$$

Comparing both sides:

$$-\Omega^2 a - 2\gamma\Omega b + \omega^2 a = F$$

$$-\Omega^2 b + 2\gamma\Omega a + \omega^2 b = 0$$

$\therefore a(\omega^2 - \Omega^2) - 2\gamma\Omega b = F$

$b(\omega^2 - \Omega^2) = -2\gamma\Omega a \Rightarrow a = \dfrac{-b(\omega^2 - \Omega^2)}{2\gamma\Omega}$

$\therefore \dfrac{-b(\omega^2 - \Omega^2)^2}{2\gamma\Omega} - 2\gamma\Omega b = F$

$\cancel{\Rightarrow -b[\omega^2 + \Omega^2 \; \omega^2 + \Omega^2)^2 = 2\gamma\Omega F}$

$\cancel{\Rightarrow b = \dfrac{2\gamma\Omega F}{(\omega^2 + \Omega^2)^2}}$

$\Rightarrow -b[(\omega^2 - \Omega^2)^2 + (2\gamma\Omega)^2] = 2F\Omega\gamma$

$\Rightarrow \boxed{b = \dfrac{-2\gamma\Omega F}{(\omega^2 - \Omega^2)^2 + (2\gamma\Omega)^2}}$ (

$$\therefore a = \frac{2\gamma\Omega F}{(\omega^2-\Omega^2)^2+(2\Omega\gamma)^2} \cdot \frac{(\omega^2-\Omega^2)}{2\gamma\Omega}$$

$$\boxed{a = \frac{F(\omega^2-\Omega^2)}{(\omega^2-\Omega^2)^2+(2\Omega\gamma)^2}}$$ ~~(Ans)~~

$\therefore$ ~~$y(x) = y_c + y_p$~~. If we write

$$y_p = a\sin(\Omega x) + b(\cos\Omega x) = R\sin(\Omega x + \delta)$$
$$= R[\sin\Omega x\cos\delta + \cos\Omega x\sin\delta]$$

Comparing both sides:

$$R\cos\delta = a$$
$$R\sin\delta = b$$
$$\Rightarrow R = \sqrt{a^2+b^2}$$

$$\boxed{R = \frac{F}{\sqrt{(\omega^2-\Omega^2)^2+(2\Omega\gamma)^2}}}$$

$$\therefore \tan\delta = \frac{b}{a} = -\frac{2\gamma\Omega}{\omega^2-\Omega^2} = \boxed{\frac{2\gamma\Omega}{\Omega^2-\omega^2} = \tan\delta}$$ (Ans)

$y(x) = y_c + y_p$. (Ans)